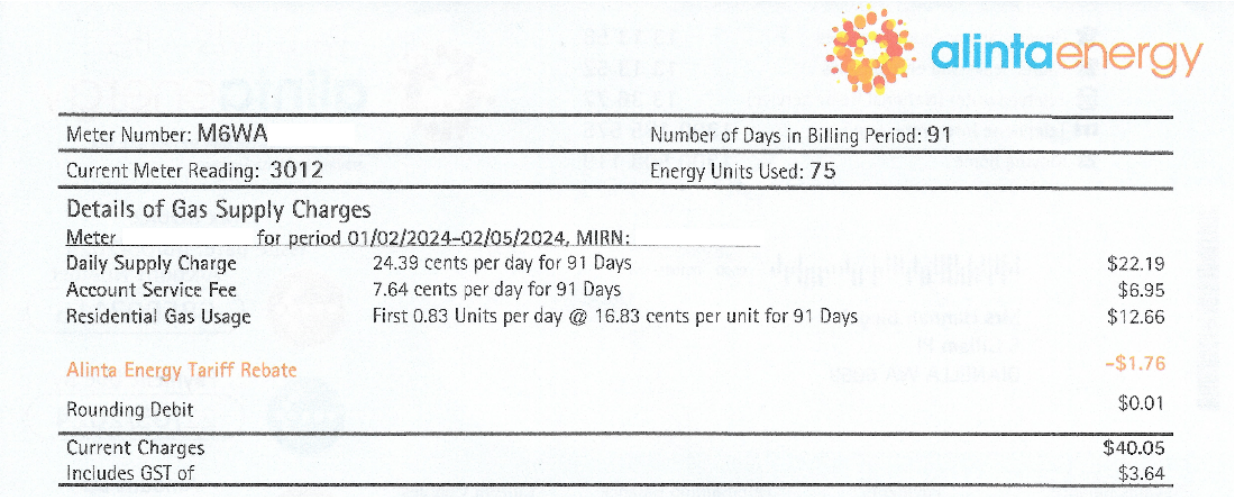
learning training. Examples include:

- A maintenance record that captures work that has been performed on a piece of equipment at some time in the past.
- An incident report or failure record that provides information about system vulnerabilities, failure modes, and recovery procedures.
- A work order that records work to be scheduled in the future.
- A utility bill, as illustrated in figure 2.1 that records a meter reading at a point in time.

Digitizing historical records is important as it allows you to understand how physical systems have performed over time, capture changes made to the physical system, and, importantly, provide the training data necessary for machine learning algorithms.



alintaenergy

Meter Number: M6WA	Number of Days in Billing Period: 91	
Current Meter Reading: 3012	Energy Units Used: 75	
Details of Gas Supply Charges		
Meter for period 01/02/2024–02/05/2024, MIRN:		
Daily Supply Charge	24.39 cents per day for 91 Days	$22.19
Account Service Fee	7.64 cents per day for 91 Days	$6.95
Residential Gas Usage	First 0.83 Units per day @ 16.83 cents per unit for 91 Days	$12.66
Alinta Energy Tariff Rebate		-$1.76
Rounding Debit		$0.01
Current Charges		$40.05
Includes GST of		$3.64

Figure 2.1 A sample gas bill as generated quarterly by my residential gas provider. The current meter reading is provided for a stated date.

TRY IT OUT: UTILITY BILL ANALYSIS

To reduce my home water and energy consumption, I must first establish the current utility usage baseline. A predictive digital twin requires observing the past, understanding the present, and forecasting the future.

Home utilities typically rely on periodic manual meter readings (quarterly for gas/water in this example) or low-resolution digital data (daily for